

G Seventy

A 70s two-oscillator mono synth: six-octave VCO, ring and noise, high- and low-pass filter knobs, auto bend and heavy drift.



Overview

G Seventy is a monophonic synth in the spirit of the mid-70s Japanese two-oscillator monos. VCO 1 gives a choice of five waves over six octaves, down to a 64' sub-bass. VCO 2 adds a second saw, a ring-modulated tone or one of three noises, and BLEND mixes the two. The sound then runs through the FILTER: a high-pass and a low-pass filter in series, each with its own knob. Move the two together and the pass band travels up and down the spectrum; pull them apart for a wide, open sound, or close them in for a thin, nasal one.

As on the originals there are no pitch or mod wheels on the panel. Instead, AUTO BEND bends the pitch into every new note and VIBRATO has its own LFO. G Seventy is unstable on purpose: DRIFT is set high by default, so the pitch, the filters and even the envelope's timing wander like an old synth warming up. Pitch bend from your keyboard still works (+/-2 semitones).

Reach for it for

- Thick, wobbly 70s leads
- Deep 32' and 64' bass
- Ring-modulated bells and metallic tones
- Wind, surf and whistle effects from the noise sources
- Brass stabs with a built-in pitch scoop

Controls

WAVE, OCTAVE and SOURCE are cycle keys: click for the next setting, right-click for the one before. The dots under each key show where you are.

VCO 1

WAVE	Triangle, saw, square, a narrow pulse, or PW (a pulse whose width is swept by the MG, the main LFO). The key shows the wave's shape, or PW.
OCTAVE	64', 32', 16', 8', 4' or 2'. 8' is standard pitch; each step down is an octave lower.

VCO 2

SOURCE	Saw (a second saw), RING (VCO 1 ring-modulated by VCO 2: clangorous, bell-like tones that change with TUNE), WHITE (white noise), PINK (pink noise, darker) or RED (red noise, a deep rumble).
TUNE	VCO 2's pitch against VCO 1, -12 to +12 semitones. A few hundredths of a semitone gives a slow beat; 7 or 12 give fifths and octaves.
BLEND	The mix: fully left is VCO 1 only, fully right VCO 2 only.

MODULATION

MG RATE	The main LFO's speed, 0.1 to 30 Hz (a triangle wave). It sweeps the PW wave and, with the MG switch, the filters. The orange light blinks with it.
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PWM	How far the MG sweeps the pulse width when VCO 1 is on PW.
VIB RATE / VIB DEPTH	Vibrato on its own LFO, separate from the MG: 1 to 12 Hz, up to 100 cents.

FILTER

HPF	The high-pass filter's cutoff, 20 Hz to 8 kHz (yellow knob).
LPF	The low-pass filter's cutoff, 60 Hz to 18 kHz (green knob). Both filters are 12 dB per octave.
RESONANCE	LOW, MID or HIGH: the peak at both cutoffs.

FILTER MOD

EG	How far the envelope opens both filters: OFF, 1/2 or 1 (about 5 octaves).
MG	How far the MG sweeps both filters: OFF, 1/2 or 1 (about 2 octaves each way).

AUTO BEND / GLIDE

BEND	Each new note starts this far from its pitch, -12 to +12 semitones, and bends into it. Negative scoops up from below, positive falls from above. 0 is off.
TIME	How long the bend takes, 10 ms to 2 s.
PORTA	Portamento: the pitch glides from one note to the next, 0 (off) to 1.5 s.

ENVELOPE

ATTACK, DECAY, SUSTAIN, RELEASE	One envelope shapes both the amp and (through the EG switch) the filters. Playing legato (overlapping notes) keeps the envelope going rather than restarting it.
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PLAYING

VELOCITY

How much key velocity changes the level, 0% (none, as on the originals) to 100%.

MASTER

DRIFT

See Drift below.

VOLUME

-24 to +6 dB.

SAT

The GFX band saturation on the output (0% by default).

OUTPUT CLIP

A soft clipper on the output (always on) catches overs at the ceiling.

Meter

The output level.

The header

PRESETS

Opens the patch list (or click the display). An asterisk after the patch number means you have changed it.

◀ / ▶

Step through the patches.

MIDI

Blinks on each note.

COLOUR

Two swatches: olive (the default) or slate. The choice is saved with the project.

S / M / L

Panel size: small, medium or large. The panel also scales to fit the plugin window.

?

Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Drift

DRIFT is heavy on G Seventy: 60% by default, scaled further than the other GFX instruments. The pitch wanders, VCO 2 drifts against VCO 1, the pulse width and

both filters move, and every new note starts slightly off tune. The envelope's timing wanders too, slowly and a little differently on every note, so attacks and releases never quite repeat. Turn it down for a steadier, more modern sound; 0% is steady.

Patches

Seventy Lead	Saw with a slightly detuned second saw, gentle vibrato and a touch of portamento.
64' Bass	A 32' square with VCO 2 an octave up: deep and round.
Filter Sweep	Two saws with the MG slowly sweeping the filters at high resonance.
Ring Bells	Ring modulation a fifth apart, long decay: bell and gong tones.
PWM Strings	A slow PWM wave with a soft attack and release.
Auto Bend Brass	Saws with the envelope fully on the filter and a short scoop up into every note.
Whistle	Mostly triangle, with a breath of pink noise through a resonant band.
Wind	Pink noise only, the filters swept slowly by the MG at high resonance.
Squelch	A 16' pulse through a nearly closed, resonant low-pass with a snappy envelope.
Theremin	A pure triangle with wide vibrato and slow portamento.
Zap Drum	A 32' triangle with white noise, dropping an octave in 70 ms: a laser-like tom.
Hollow Pipe	A square with VCO 2 an octave up through a narrow band.

Tips

- Set HPF and LPF close together with RESONANCE on HIGH, then turn them as a pair for a talking, vocal sweep.
- For ring-mod effects, try TUNE settings that are not whole semitones.
- AUTO BEND with a short TIME and -1 to -3 semitones gives brass-like scoops; long times give sci-fi swoops.
- With OCTAVE on 64', keep the HPF low or the fundamental disappears.

File: gfx-g-seventy.jsfx